

Lepton Masses from Rope Topology

A Koide-Phase Model: Numerical Success, and Its Pinned Failure Mode

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Abstract

The rope framework suggests a topological origin for the charged-lepton mass ratios through a Koide-type phase relation with coefficient $(3+\Phi) = 4.618034$, where Φ is the golden ratio. Used with the measured Weinberg angle, the relation reproduces the lepton mass ratios to within about one percent: $\mu/e = 205.8$ (measured 206.8) and $\tau/\mu = 16.82$ (measured 16.82). We report this success and, with equal prominence, its failure mode: with the rope model's own predicted Weinberg angle the ratios are badly wrong. The result is a numerically striking CONJECTURE conditional on a measured input, not a derivation. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants; the lepton-ratio caveat below is pinned in the test suite.

Scope of this paper

CLAIM: a Koide-phase relation with coefficient $(3+\Phi)=4.618$ reproduces the charged-lepton mass ratios to ~1% WHEN the measured $\sin^2\theta_W$ is used.

NOT CLAIMED: a first-principles derivation. With the model's own $\sin^2\theta_W$ the ratios fail; the success is conditional on a measured input. Status: Conjecture.

1. The Koide Phase and Its Coefficient

The empirical Koide relation among charged-lepton masses is reproduced in a phase parametrisation whose coefficient the rope framework computes as $(3+\Phi) = 4.618034$, with Φ the golden ratio 1.618034. This coefficient is the model's structural contribution.

2. Numerical Result

Evaluated with the measured Weinberg angle $\sin^2\theta_W = 0.23122$, the relation yields:

Ratio	Rope (measured θ_W)	Experiment	Agreement
μ/e	205.8	206.8	~0.5%
τ/μ	16.82	16.82	~0.04%
τ/e	3462.7	3477.2	~0.4%

The pinned honest caveat. The lepton mass ratios come out close to experiment ONLY when the measured Weinberg angle $\sin^2\theta_W = 0.23122$ is used as input. With the rope model's OWN predicted value $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ — itself about 1.9% from the measured value — the ratios fail badly (μ/e jumps from ~ 206 to ~ 1605). The mass-ratio success is therefore CONDITIONAL on an externally measured input the model does not itself reproduce accurately. This is a conjecture with a known failure mode, not a derivation, and it is recorded as such in the test suite.

Status

Koide coefficient $(3+\Phi) = 4.618$ — Derived (from golden-ratio structure).

Lepton ratios to $\sim 1\%$ with MEASURED θ_W — Modeled (Conjecture).

Ratios with the model's OWN θ_W — Failed ($\mu/e \sim 1605$); pinned in tests.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **PM-001 [Conjecture]:** Lepton mass ratios via Koide phase

- **PM-002 [Failed]:** Lepton ratios with the model's own Weinberg angle

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.